Preparation and Analysis of Potassium Tris(Oxalato)Ferrate(III)Trihydrate

A General Chemistry Experiment

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The preparation of potassium tris(oxalato)ferrate(III) trihydrate as a laboratory experiment has been suggested for both first-year general chemistry (*1*, *2*) and graduate-level work (*3*). At California State University, Fullerton, a modified experiment provides general chemistry students with an integrated experience in preparing and analyzing a compound. The preparation is simple yet requires reasonable care to give a high yield of product, while the analysis combines gravimetric and titrimetric methods and includes a unique and colorful photoredox reaction. The experiment is allocated a total of five 3-hour laboratory periods: one for the synthesis, one for determination of water of hydration, and three for the determinations of iron and oxalate, which are carried out simultaneously.

Students prepare the compound as described in the literature (*2*), with one modification: the final product is rinsed with two aliquots of cold methanol to remove excess water. Following the rinses, the crystals are spread on filter paper, covered, and air dried. In a class of 30 students, the average yield was 82% with a standard deviation of 14%. Excessively high apparent yields can result if the product is inadequately dried, while low yields are typically caused by using too large a volume of distilled water in the recrystallization step.

The empirical formula of the product is determined in three steps: water is determined gravimetrically; oxalate is titrated with standardized $KMnO_4$; and iron is photolytically reduced to ferrous and precipitated as ferrous oxalate. Potassium content is obtained by difference.

The entire yield of product is first crushed to a powder using a mortar and pestle, accurately weighed, and then dried to constant weight in an oven at 110 °C. (One hour is usually sufficient.) It is convenient for the students to divide their sample into two portions, a smaller 0.5-g sample on which the accurate gravimetric procedure is done and the remainder of the product, which is simply dried for 1.5–2.0 hours. Following dehydration, the compound is stored in brown bottles in desiccators, under which conditions it is stable indefinitely.

The anhydrous salt is analyzed for iron by dissolving a weighed sample of about 1 g in 15 ml of 10% (v/v) acetic acid in a test tube. This solution is then exposed to sunlight whereupon the classic ferrioxalate photoredox reaction (4) occurs:

$$2\ Fe^{3+} + C_2O_4{}^{2-} \xrightarrow{h\nu} 2\ Fe^{2+} + 2CO_2$$

Since the primary species in solution is $Fe(C_2O_4)^+$ and iron(II) is very sparingly soluble in oxalate solution, the net reaction is

$$2\ [Fe(C_2O_4)]^+ + (C_2O_4)^{2-} + 4\ H_2O \xrightarrow{h\nu} 2\ FeC_2O_4 \cdot 2\ H_2O + 2\ CO_2$$

In direct sunlight, this reaction goes to completion in about 2 hr, and even in diffuse sunlight it is complete within 6–8 hr. To insure completeness, the test tubes are placed on a windowsill and left until the succeeding laboratory period.

Initially, the acid solution is emerald green and the ferrioxalate is incompletely dissolved. As the reaction proceeds, the remaining solid dissolves, the solution gradually bleaches, effervescence is observed as CO_2 is evolved, and canary yellow ferrous oxalate precipitates out, leaving a virtually colorless supernatant solution.

The ferrous oxalate is isolated by filtration and rinsing with acetone and is then weighed as $FeC_2O_4{\cdot}2H_2O$. The most convenient method is using a previously tared filter crucible; if filter crucibles are not readily available, a Büchner funnel fitted with preweighed, oversize filter paper fitted into a cup is also satisfactory. In either case, the precipitate is washed once with 10% acetic acid and twice with acetone, following which it is dried under suction for 10 min prior to final weighing. Use of the filter paper cup gives larger errors from water adsorbed on the filter paper. Oven drying the filter paper at 60°C for several minutes or storing filter paper and precipitate until the next laboratory period reduces this error.

Finally, the oxalate content of the compound is determined by titrating weighed samples dissolved in hot sulfuric acid with 0.1 *N* standard potassium permanganate (*5*). The standard permanganate can be provided either as a stock solution of known normality or students may be required to standardize their permanganate against sodium oxalate.

Results are summarized in the table for a typical group of general chemistry students. Student results for the water of hydration tend to be systematically high by about 10%, because of failure to remove all extraneous water before beginning the analysis. Careful students can avoid this error: one-quarter of our student sample obtained a result within 2% of the correct value. The photolytic/gravimetric analysis for iron yields average results that are about 4% too low, due to the sparing solubility of ferrous oxalate in water and the difficulty of quantitatively transferring the precipitate. Nonetheless, 40% of our student sample obtained results within 2% of the true value. The oxalate titration by permanganate is quite

Analysis of $K_3Fe(C_2O_4)_3{\cdot}3H_2O$
Summary of Student Results

	H_2O	Fe^{3+}	$C_2O_4{}^{2-}$	K^+
mmoles/g, actual [a]	6.87	2.29	6.87	6.87
mmoles/g, students [b]	7.44	2.21	6.72	7.47
standard deviation [c]	0.81	0.10	0.39	0.82
stoichiometric ratio [d]	3.37	1.00	3.04	3.38
% within 2% [e]	25	40	50	30

[a] Expressed as millimoles per gram of anhydrous salt.
[b] Averaged student results, 2 laboratory classes totalling 30 students.
[c] Standard deviation in student results, mmoles/g.
[d] Average student result assuming 1 Fe per formula unit.
[e] Percentage of students obtaining a result within 2% of the correct value.

accurate even when, as here, the students are asked to perform their own standardization. The potassium result, being determined by difference, is susceptible to cumulative errors. This possibility is reduced by calculating potassium for the anhydrous salt, thereby eliminating error contributions from the water of hydration.

This experiment offers several advantages for use in general chemistry. It integrates preparative and analytic techniques. It utilizes a photochemical process which can excite student interest both from its visual impact (students can see photochemistry taking place in the test tube) and as an introduction to photoinduced processes. It rewards the careful student with accurate results. Finally, it currently costs less than $0.20 per student per laboratory period in expendable materials.

Literature Cited

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